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LOADING PORT STRUCTURE, SUBSTRATE PROCESSING APPARATUS, SUBSTRATE PROCESSING METHOD AND METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE

Abstract

It is possible to suppress a tilt of a substrate container when a lid of the substrate container is opened. There is provided a technique that includes: a frame separating inside and outside of a local clean environment, and including: a first surface outside the local clean environment airtightly contacting an opening of a substrate container; and an entrance for the substrate container; and a latching structure switching between an engaged state where it engages with a flange around the opening and a disengaged state where a movement of the substrate container is unrestricted in an up-down direction and in a front-rear direction. The latching structure includes: heads engaging with recesses on each side of the flange; and drivers moving the heads in a direction parallel to the first surface, the heads having a contact portion generating a pressing force for the flange against the first surface.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION
[0001] This non-provisional U.S. patent application is based on and claims priority under 35 U.S.C. § 119 of Japanese Patent Application No. 2024-031649 filed on Mar. 1, 2024, in the Japanese Patent Office, the entire contents of which are hereby incorporated by reference.
BACKGROUND
1. Field
[0002] A technique of the present disclosure relates to a loading port structure, a substrate processing apparatus, a substrate processing method and a method of manufacturing a semiconductor device.
2. Related Art
[0003] According to some related arts, as a part of a manufacturing process of a semiconductor device, a process of opening a lid of a substrate container placed on a loading port structure and transferring a substrate accommodated in the substrate container may be performed.
SUMMARY
[0004] According to the present disclosure, there is provided a technique capable of suppressing a tilt of a substrate container when a lid of the substrate container is opened.
[0005] According to an embodiment of the present disclosure, there is provided a technique that includes: a frame constituting a part of a wall configured to separate an inside and an outside of a local clean environment, and including: a first surface provided outside the local clean environment and configured to be capable of making an airtight contact with an opening of a substrate container; and an entrance through which a substrate accommodated in the substrate container is capable of passing; and a latching structure provided adjacent to the first surface and configured to be capable of being switched between an engaged state in which the latching structure engages with a flange provided around the opening and a disengaged state in which the latching structure does not engage with the flange so that a movement of the substrate container is unrestricted in an up-down direction and in a front-rear direction, wherein the latching structure includes: a pair of heads configured to respectively engage with at least two recesses formed on each side of the flange; and a pair of drivers configured to respectively move the pair of heads in a direction substantially parallel to the first surface, wherein each of the pair of heads is provided with a contact portion configured to generate a force of pressing the flange against the first surface by transitioning the latching structure from the disengaged state to the engaged state, and wherein the contact portion includes a tapered surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0006] FIG. **1** is a diagram schematically illustrating an exemplary perspective view of a substrate processing apparatus according to one or more embodiments of the present disclosure.
[0007] FIG. **2** is a diagram schematically illustrating a vertical cross-section of a locking structure of a loading port structure according to the embodiments of the present disclosure.

[0008] FIG. **3** is a diagram schematically illustrating an exemplary perspective view of a substrate container according to the embodiments of the present disclosure.
[0009] FIG. **4**A is a diagram schematically illustrating a state in which the substrate container is fixed to a frame of the loading port structure by a latching structure according to the embodiments of the present disclosure, and FIG. **4**B is a diagram schematically illustrating a state in which the substrate container is released from the frame of the loading port structure by the latching structure shown in FIG. **4**A.
[0010] FIG. **5** is a diagram schematically illustrating the latching structure when viewed along a Y-axis direction of FIG. **4**A.
[0011] FIG. **6** is a flow chart schematically illustrating an exemplary process flow of a substrate processing performed by the substrate processing apparatus according to the embodiments of the present disclosure.
[0012] FIG. **7**A is a diagram schematically illustrating a state in which the substrate container is fixed to the frame of the loading port structure by a latching structure according to a first modified example of the present disclosure, and FIG. **7**B is a diagram schematically illustrating a state in which the substrate container is released from the frame of the loading port structure by the latching structure shown in FIG. **7**A.
[0013] FIG. **8**A is a diagram schematically illustrating a state in which the substrate container is fixed to the frame of the loading port structure by a latching structure according to a second modified example of the present disclosure, and FIG. **8**B is a diagram schematically illustrating a state in which the substrate container is released from the frame of the loading port structure by the latching structure shown in FIG. **8**A.
[0014] FIG. **9** is a diagram schematically illustrating a partial cross-section of the latching structure according to the second modified example.
[0015] FIG. **10** is a diagram schematically illustrating a latching structure according to a third modified example of the present disclosure.
[0016] FIG. **11** is a diagram schematically illustrating a latching structure according to a fourth modified example of the present disclosure.
[0017] FIG. **12**A is a diagram schematically illustrating a cross-section of the latching structure according to the fourth modified example, taken along a line A-A shown in FIG. **11**, when the latching structure according to the fourth modified example is in contact with a flange of the substrate container, and FIG. **12**B is a diagram schematically illustrating the cross-section of the latching structure according to the fourth modified example, taken along the line A-A shown in FIG. **11**, when the latching structure according to the fourth modified example is released from contact with the flange of the substrate container.

DETAILED DESCRIPTION

[0018] Hereinafter, one or more embodiments (also simply referred to as “embodiments”) according to the technique of the present disclosure will be described mainly with reference to FIGS. **1** to **12**B. Further, the drawings used in the following descriptions are all schematic. For example, a relationship between dimensions of each component and a ratio of each component shown in the drawing may not always match the actual ones. Further, even between the drawings, the relationship between the dimensions of each component and the ratio of each component may not always match. In addition, the same or similar reference numerals represent the same or similar components in the drawings. Thus, each component is described with reference to the drawing in which it first appears, and redundant descriptions related thereto will be omitted unless particularly necessary. In addition, the technique of the present disclosure is not limited to the embodiments described below. That is, the technique of the present disclosure may be modified in various ways without departing from the scope thereof.

(1) OVERALL CONFIGURATION OF SUBSTRATE PROCESSING APPARATUS

[0019] FIG. **1** is a diagram schematically illustrating an exemplary perspective view of a substrate processing apparatus **1** according to the embodiments of the present disclosure.

[0020] For convenience of explanation, three mutually orthogonal axes (that is, an X-axis, a Y-axis, and a Z-axis) are shown in FIG. **1**. The X-axis is an axis along a first direction in a horizontal direction (that is, a front-rear direction of the substrate processing apparatus **1**), the Y-axis is an axis along a second direction (which is perpendicular to the X-axis) in the horizontal direction (that is, a left-right direction of the substrate processing apparatus **1**), and the Z-axis is an axis along a vertical direction (that is, an up-down direction of the substrate processing apparatus **1**).

[0021] As shown in FIG. **1**, the substrate processing apparatus **1** includes a housing **2**. An opening for performing a maintenance operation is provided at a lower portion of a front wall **3** of the housing **2**. The opening can be opened or closed by a front maintenance door **5**.

[0022] A pod loading/unloading port is provided at the front wall **3** of the housing **2** so as to communicate with an inside and an outside of the housing **2**. The pod loading/unloading port can be opened or closed by a front shutter (which is a pod loading/unloading port opening/closing structure) **7**. A pod transfer table **8** is provided in front of the pod loading/unloading port. The pod transfer table **8** is configured such that a pod **9** serving as a substrate container (also referred as a FOUP (Front Opening Unified Pod)) is aligned while placed on the pod transfer table **8**.

[0023] For example, the pod **9** is configured as a sealed type substrate transfer container. A plurality of wafers **18** can be accommodated in the pod **9**. Hereinafter, each of the wafers **18** may also be referred to as a “wafer **18**” which serves as a substrate. The wafer **18** can be transferred while accommodated in the pod **9**. For example, the pod **9** can be transferred (loaded) into and placed on the pod transfer table **8** by an in-process transfer apparatus (not shown) and can be transferred (unloaded) from the pod transfer table **8** by the in-process transfer apparatus.

[0024] A rotatable pod shelf (which is a pod storage shelf) **11** is provided in the housing **2** to be located over a substantially center portion of the housing **2** in an X-axis direction. The rotatable pod shelf **11** is configured such that a plurality of pods including the pod **9** can be stored (or placed) on the rotatable pod shelf **11**. Hereinafter, the plurality of pods including the pod **9** may also be simply referred to as “pods **9**”.

[0025] A pod opener (which is a pod lid attaching/detaching structure) **14** is provided below the rotatable pod shelf **11**. The pod opener **14** is provided with a configuration capable of attaching and detaching a lid of the pod **9**.

[0026] A pod transfer structure (which is a container transfer structure) **15** is provided among the pod transfer table **8**, the rotatable pod shelf **11** and the pod opener **14**. The pod transfer structure **15** is configured such that the pod **9** can be elevated and lowered in a Z-axis direction (also referred to as the “up-down direction”) while being supported by the pod transfer structure **15**, and can be moved forward and backward in the horizontal direction (that is, in a Y-axis direction (also referred to as the “left-right direction”) and the X-axis direction (also referred to as the “front-rear direction”)) while being supported by the pod transfer structure **15**. The pod transfer structure **15** is further configured such that the pod **9** can be transferred among the pod transfer table **8**, the rotatable pod shelf **11** and the pod opener **14**.

[0027] A sub-housing **16** is provided below the substantially center portion of the housing **2** in the X-axis direction to extend toward a rear end of the substrate processing apparatus **1**. A pair of wafer loading/unloading ports through which the wafer **18** is loaded into or unloaded out of the sub-housing **16** is provided at a front wall **17** of the sub-housing **16**. The pair of wafer loading/unloading ports is arranged vertically in two stages. A pair of pod openers (including the pod opener **14**) is provided at the pair of wafer loading/unloading ports (which are openings), respectively. For example, an upper pod opener and a lower pod opener may be provided as the pair of pod openers. Hereinafter, the upper pod opener may also be referred to as an “upper pod opener **14**” or the “pod opener **14**”, and the lower pod opener may also be referred to as a “lower pod opener **14**” or the “pod opener **14**”.

[0028] A loading port structure **21** configured to transfer the wafer **18** in the pod **9** is provided on the front wall **17**. The loading port structure **21** includes a mounting table (which is a placement table) **22** where the pod **9** is placed thereon. By attaching or detaching the lid of the pod **9** placed on the mounting table **22** using the pod opener **14**, the wafer **18** can be transferred to and from the pod **9**.

[0029] The sub-housing **16** defines a transfer chamber **23** fluidically isolated from a space (hereinafter, also referred to as a "pod transfer space") in which the pod transfer structure **15** or the rotatable pod shelf **11** is provided. A wafer transfer structure (which is a transfer structure) **24** is provided in the transfer chamber **23** (hereinafter, also referred to as a "substrate transfer space"). The wafer transfer structure **24** is configured to transfer (load) the wafer **18** into and to transfer (unload) the wafer **18** out of a boat (which is a substrate retainer) **26**.

[0030] In the transfer chamber **23**, a standby space **27** where the boat **26** is accommodated and in standby is provided, and a process furnace **28** such as a vertical type process furnace is provided above the standby space **27**. A process chamber **29** is provided inside the process furnace **28**, and a lower end portion of the process chamber **29** is configured as a furnace opening. The furnace opening is opened or closed by a furnace opening shutter. The process furnace **28** serves as an example of a process vessel in which the wafer **18** is processed.

[0031] A boat elevator (which is a substrate retainer elevating structure) **32** configured to elevate and lower the boat **26** is provided below the process furnace **28**. A seal cap (which is a lid) **34** is horizontally attached to the boat elevator **32**. The seal cap **34** is configured such that the boat **26** can be vertically supported by the seal cap **34**, and such that the furnace opening can be airtightly closed by the seal cap **34** while the boat **26** is loaded into the process chamber **29**.

[0032] The boat **26** is configured such that a plurality of wafers (for example, from 50 wafers to 125 wafers) including the wafer **18** are supported on the boat **26** in a horizontal orientation and in a multistage manner with their centers aligned with one another. Further, in the present specification, a notation of a numerical range such as "from 50 wafers to 125 wafers" means that a lower limit and an upper limit are contained in the numerical range. Therefore, for example, a numerical range "from 50 wafers to 125 wafers" means a range equal to or higher than 50 wafers and equal to or less than 125 wafers. The same also applies to other numerical ranges described in the present specification.

[0033] A clean air supplier (which is a clean air supply structure or a clean air supply system) **35** is arranged at a location facing the boat elevator **32**. The clean air supplier **35** is constituted by a supply fan and a dustproof filter so as to supply clean air such as an inert gas and a clean atmosphere.

[0034] As shown in FIG. **1**, the substrate processing apparatus **1** includes a controller **100**. The controller **100** is configured to be capable of controlling the substrate processing apparatus **1**. The controller **100** may be embedded in the substrate processing apparatus **1**, or may be provided so as to be capable of accessing to the substrate processing apparatus **1** from an outside of the substrate processing apparatus **1**. While the present embodiments will be described by way of an example in which the controller **100** according to the present embodiments is applied to the substrate processing apparatus **1**, the controller **100** may control an apparatus other than the substrate processing apparatus **1**.

(2) CONFIGURATION OF LOADING PORT STRUCTURE

[0035] Subsequently, a configuration of the loading port structure **21** will be described. FIG. **2** is a diagram schematically illustrating a vertical cross-section of a locking structure **41** of the loading port structure **21**. FIG. **3** is a diagram schematically illustrating an exemplary perspective view of the pod **9**. In FIG. **2**, in order to explain a configuration of the locking structure **41** of the loading port structure **21**, a latching structure **43** (which brings the pod **9** into contact with a frame **40** (which is a part of the front wall **17**)) is not illustrated in detail. In addition, the controller **100** is electrically connected to the locking structure **41** and the latching structure **43**. That is, the controller **100** is configured to be capable of controlling components of the locking structure **41** and components of the latching structure **43** such that a desired operation can be performed at a desired timing.

[0036] For example, the loading port structure **21** includes: the frame **40** constituting a part of the front wall **17** configured to separate an inside and an outside of a local clean environment (also referred to as a "mini environment"); the mounting table **22** whose one end is fixed to the frame **40**; the locking structure **41** configured to position the pod **9** placed on the mounting table **22**; and the latching structure **43** (described in detail later) configured to press the pod **9** placed on the mounting table **22** against the frame **40**.

[0037] That is, the frame **40** is configured to separate the transfer chamber **23** (which is the local clean environment) from the pod transfer space (which is outside the local clean environment). For example, the frame **40** includes: a surface **40***a* serving as a first surface provided adjacent to the pod transfer space; a surface **40***b* serving as a second surface provided adjacent to the transfer chamber **23**; and an opening **40***c* through which the wafer **18** accommodated in the pod **9** can pass (or can be transferred). The opening **40***c* may also be referred to as an "entrance". In other words, the opening **40***c* through which the wafer **18** can pass is provided in the front wall **17** configured to separate the inside and the outside of the local clean environment.

[0038] The mounting table **22** is configured such that the pod **9** can be placed thereon through the pod transfer space.

[0039] For example, the locking structure **41** is constituted by: a support table **45** provided on the mounting table **22** and configured such that the pod **9** can be placed thereon; a plurality of guide structures **46** configured to slidably move the support table **45** in the X-axis direction; a plurality of pins **48** respectively fit into a plurality of recesses (concave portions) **47** provided in a bottom surface of the pod **9** so as to position the pod **9** on the support table **45**; and an engaging structure **50** configured to be rotated so as to engage with an engagement groove (also referred to as an "engagement recess") **49** provided in the bottom surface of the pod **9**. Hereinafter, each of the plurality of guide structures **46** may also be referred to as a "guide structure **46**".

[0040] The engaging structure **50** is configured to be capable of being rotated by a driver (which is a driving structure) **51**, and is further configured to be capable of being rotated between a position where the engaging structure **50** engages with the engagement groove **49** and another position where an engagement between the engaging structure **50** and the engagement groove **49** is released. The support table **45** is configured to be movable in a direction substantially perpendicularly (vertically) to the surface **40***a* as the guide structure **46** moves on the mounting table **22** in the X-axis direction. In a manner described above, the pod **9** placed on the support table **45** of the loading port structure **21** is positioned and fixed on the mounting table **22**.

[0041] The latching structure **43** is configured such that a flange **54** of the pod **9** placed on the mounting table **22** makes an airtight contact with the frame **40** via a seal **42**. In other words, the latching structure **43** is configured such that an opening **44** of the pod **9** placed on the mounting table **22** can make an airtight contact with the surface **40***a* around the opening **40***c*.

[0042] The pod **9** is configured such that the wafers **18** (for example, 25 wafers) can be accommodated in the pod **9** while being arranged substantially horizontally in a multistage manner. The opening **44** through which the wafer **18** is inserted or removed (that is, loaded or unloaded) is provided in a first end surface of the pod **9**, and a lid **52** corresponding to the opening **44** is attached to the opening **44** such that the lid **52** can be attached to or detached from the opening **44**. As shown in FIG. **3**, for example, the lid **52** is provided with two keyholes **53**. By locking or unlocking the keyholes **53** with the pod opener **14**, the lid **52** can be attached to or detached from the opening **44**. An inside (inner portion) of the pod **9** is configured to be substantially airtight by the lid **52**.

[0043] The flange **54** protruding from each peripheral surface of the pod **9** toward an outer periphery thereof is provided around the opening **44** of the pod **9**. For example, two recesses **55** are provided on each of side surfaces of the flange **54** with a predetermined interval therebetween in the Z-axis direction. Hereinafter, each of the recesses **55** may also be referred to as a "recess **55**". A space (which resembles an inverted truncated pyramid) is provided inside the recess **55**. Thereby, a side surface of the recess **55** is slightly inclined rather than perpendicular to a depth direction. The recesses **55** are not limited to such a configuration where they are provided at two locations on each side surface of the flange **54**. For example, instead of the two recesses **55**, one or more recesses may be provided at one or more locations on each side surface of the flange **54**. In addition, the flange **54** may be configured as a lattice-like rib structure protruding outward, and each of the recesses **55** may correspond to one of concave portions interposed between ribs of the lattice-like rib structure.

[0044] In addition, a handle (which is of a flange shape) **56** capable of being gripped by a transfer robot is provided on an upper surface of the pod **9**.

[0045] Subsequently, a configuration of the latching structure **43** will be described. FIG. **4**A is a diagram schematically illustrating a state in which the pod **9** is fixed to the surface **40***a* of the frame **40** by the latching structure **43**. FIG. **4**B is a diagram schematically illustrating a state in which the pod **9** is released from the surface **40***a* of the frame **40** by the latching structure **43**. FIG. **5** is a diagram schematically illustrating the latching structure **43** when viewed along the Y-axis direction of FIG. **4**A. In addition, in FIG. **5**, the pod **9** is shown by a two-dot chain line to explain the latching structure **43** in detail.

[0046] The latching structure **43** is fixed to the surface **40***a* of the frame **40** by a support (which is a support structure) **60** of a plate shape. That is, the latching structure **43** is provided at the surface **40***a* outside the local clean environment, and not adjacent to other surfaces of the frame **40** than the surface **40***a*. Therefore, the latching structure **43** can be arranged not to interfere with components such as the pod opener **14** and a mapping sensor, and can be easily attached to the substrate processing apparatus **1**.

[0047] For example, the latching structure **43** includes: a pair of heads **62** respectively engaging with at least two recesses **55** formed on each side surface of the flange **54**; and a pair of drivers (driving structures) **63** configured to respectively drive the pair of heads **62** to move in a direction substantially parallel to the surface **40***a*. Configurations of the pair of heads **62** are substantially the same with each other and configurations of the pair of drivers **63** are substantially the same with each other. In addition, the pair of heads **62** (and the pair of drivers **63**) is arranged symmetrically with respect to a plane and is located at positions facing each other with the opening **40***c* of the surface **40***a* interposed therebetween. Hereinafter, each of the pair of heads **62** may also be referred to as a “head **62**”, and each of the pair of drivers **63** may also be referred to as a “driver **63**”. Therefore, in the following description, the present embodiments will be described using the head **62** and the driver **63** engaging with the recess **55** provided on each side surface of the pod **9**.

[0048] The head **62** is provided with a contact portion **64** at one end thereof. The contact portion **64** comes into contact with the side surface of the recess **55** and generates a force of pressing the flange **54** against the surface **40***a*. The contact portion **64** is provided with a first tapered surface **64***a* inclined relative to a direction along which the pod **9** moves to a docking position at which the pod **9** is pressed against the surface **40***a*. A pivot **65** is inserted into an axial hole provided at approximately a center of the head **62**, and the head **62** is provided so as to be rotatable around the pivot **65**. The pivot **65** is fixed to the support **60** so as to be substantially perpendicular to the surface **40***a*. The contact portion **64** is provided with the first tapered surface **64***a* and a second tapered surface **64***b*. The first tapered surface **64***a* is chamfered with a surface substantially perpendicular to the surface **40***a* at a corner (front end) of the contact portion **64** that comes into the earliest contact with the flange **54** when the head **62** is rotated. The second tapered surface **64***b* is chamfered with a surface slightly inclined with respect to the surface **40***a*. A tilt (inclination) of this surface is set such that it moves away from the surface **40***a* as it approaches the front end of the head **62**. For example, an angle of the surface is set within a range from 1° to 40°.

[0049] Here, a position where the pod **9** is pressed against the surface **40***a* by a movement of the guide structure **46** in the loading port structure **21** may also be referred to as the “docking position”, and a position where the pod **9** is separated from the surface **40***a* may also be referred to as an "undocking position".

[0050] For example, the driver **63** is constituted by: a cylinder **66** capable of expanding and contracting on a plane substantially parallel to the flange **54** to be locked (or latched) with; and a bias structure **67** configured to bias the head **62** toward the surface **40***a*. As the cylinder **66**, for example, an air cylinder capable of expanding and contracting due to a change in an air pressure may be used. In addition, as the bias structure **67**, for example, a coil spring capable of being arranged along an outer periphery of the pivot **65** may be used. Thereby, the latching structure **43** can directly set an appropriate pressing force capable of maintaining the pod **9** airtight without interfering with a smooth locking operation (latching operation).

[0051] The cylinder **66** is rotatably fixed to the support **60** by a pin **68** provided on one end (first end) of the cylinder **66**. In addition, the cylinder **66** rotatably supports the head **62** by a pin **69** provided on the other end (second end) of the cylinder **66** and opposite to the pin **68**. That is, the cylinder **66** is configured to move in a direction without a component in the X-axis direction, to expand and contract substantially horizontally with respect to the surface **40***a*. Further, the cylinder **66** is rotated with respect to the support **60** by the pin **68**, and the cylinder **66** is further configured to rotate the head **62** by the pin **69**.

[0052] The latching structure **43** is configured to be capable of being switched between an engaged state in which the latching structure **43** engages with the recess **55** of the flange **54** as shown in FIG. **4**A and a disengaged state in which the latching structure **43** does not engage with the recess **55** of the flange **54** so that a movement of the pod **9** is unrestricted in the X-axis direction, the Y-axis direction, and the Z-axis direction as shown in FIG. **4**B.

[0053] A maximum length of the latching structure **43** in the Y-axis direction (also referred to as a "width of the latching structure **43**") is set to a length that does not interfere with the pod **9** when the pod **9** is transferred to the loading port structure **21**, for example, 45 mm or less. In addition, a maximum length of the latching structure **43** in the X-axis direction from the surface **40***a* (also referred to as a "thickness protruding from the surface **40***a*") is set to a length that does not interfere with the pod **9** when the pod **9** is transferred to the loading port structure **21**, and is set to be smaller than a distance between the flange **54** and the surface **40***a* at the undocking position where the locking structure **41** is released and the pod **9** is separated from the surface **40***a*.

[0054] The head **62** is rotated about the pivot **65** by the driver **63**. More specifically, the pair of heads **62** is rotated by the pair of drivers **63**. Thereby, the first tapered surface **64***a* of the contact portion **64** slides on the flange **54** when the head **62** moves to the docking position. As a result, the head **62** is pushed in a direction opposite to a direction along which the flange **54** is pressed.

[0055] The pair of heads **62** is configured to transition the latching structure **43** between the engaged state and the disengaged state mentioned above by the pair of drivers **63** in a synchronized manner with an accuracy enough to prevent a positional deviation in the Y-axis direction of the pod **9**. The pair of heads **62** generates the force of pressing the flange **54** against the surface **40***a* by transitioning the latching structure **43** from the disengaged state to the engaged state.

[0056] For example, in the loading port structure **21**, when the lid **52** of the pod **9** is opened, an oxygen concentration in the transfer chamber **23** increases. Therefore, when the wafer **18** is being transferred, a supply of the inert gas into the transfer chamber **23** is increased to exhaust oxygen. As a result, a pressure (inner pressure) of the transfer chamber **23**, which is inside the local clean environment, is 200 Pa or more higher than the outside of the local clean environment, and the pod **9** may come off the loading port structure **21** when the lid **52** of the pod **9** is opened. In addition, when the pod **9** comes off the loading port structure **21**, a gap may be created. As a result, an atmosphere inside and outside the local clean environment may leak. According to the embodiments of the present disclosure, by contacting an upper side surface of the pod **9** with the frame **40** (which is part of the front wall **17**) in addition to the bottom surface of the pod **9**, when the lid **52** of the pod **9** is opened, it is possible to prevent a tilt of the pod **9** in the X-axis direction, the Y-axis direction and the Z-axis direction from occurring even when the pressure inside the local clean environment is 200 Pa or more higher than the pressure outside the local clean environment.

(3) SUBSTRATE PROCESSING

[0057] Subsequently, as a part of a manufacturing process of a semiconductor, a substrate processing of processing the wafer **18** by using the substrate processing apparatus **1** mentioned above will be described with reference to FIG. **6**. In the following description, operations of components constituting the substrate processing apparatus **1** are controlled by the controller **100**.

[0058] First, when the pod **9** is transferred to the pod transfer table **8**, the pod loading/unloading port is opened by the front shutter **7**. Then, the pod **9** placed on the pod transfer table **8** is transferred (loaded) into the housing **2** by the pod transfer structure **15**, and is placed on the rotatable pod shelf **11**.

[0059] The pod **9** is temporarily stored in the rotatable pod shelf **11**. Then, the pod **9** is transferred to one of the pod openers **14** (that is, one of the upper pod opener **14** and the lower pod opener **14**) and then to the loading port structure **21** by the pod transfer structure **15**. Alternatively, the pod **9** may be transferred directly from the pod transfer table **8** to the loading port structure **21**. When the pod **9** is being transferred, the opening **40***c* is closed by the pod opener **14**, and the transfer chamber **23** is supplied with and filled with the clean air.

<Docking Step S1>

[0060] After the pod **9** is transferred to the loading port structure **21**, the pod **9** is docked by pressing an end surface of the flange **54** (that is, the end surface of the flange **54** adjacent to the opening **44**) against the surface **40***a* of an edge of the opening **40***c* of the frame **40** by the locking structure **41** and the latching structure **43**. Then, the lid **52** is detached by the pod opener **14**, and the opening **44** is opened to communicate with a local clean space (that is, the local clean environment).

<Substrate Loading Step S2>

[0061] The wafer **18** is taken out from the pod **9** by the wafer transfer structure **24**. Then, the wafer **18** is aligned by a notch aligner (not shown). Then, by the wafer transfer structure **24**, the wafer **18** is transferred (or loaded) into the standby space **27** provided in a rear region of the transfer chamber **23**, and loaded (or charged) into the boat **26**.

[0062] After the wafer **18** is charged into the boat **26**, the wafer transfer structure **24** then returns to the pod **9** and transfers a subsequent wafer among the wafers **18** from the pod **9** into the boat **26**. When a predetermined number of wafers among the wafers **18** are charged into the boat **26**, the furnace opening of the process furnace **28** closed by the furnace opening shutter is opened. Subsequently, the boat **26** is elevated by the boat elevator **32** such that the boat **26** is loaded (inserted) into the process chamber **29**, and the furnace opening is airtightly closed by the seal cap **34**.

<Substrate Processing Step S3>

[0063] The process chamber **29** is supplied with the inert gas and is vacuum-exhausted such that a pressure (inner pressure) of the process chamber **29** reaches and is maintained at a desired pressure. In addition, the process chamber **29** is heated to a predetermined temperature such that a desired temperature distribution in the process chamber **29** can be obtained.

[0064] Then, a process gas whose flow rate is controlled to a predetermined flow rate is supplied by a gas supply structure (which is a gas supplier or a gas supply system) (not shown), and the process gas comes into contact with a surface of the wafer **18** while flowing through the process chamber **29**. Thereby, a predetermined processing is performed on the surface of the wafer **18**. Further, the process gas after a reaction of the predetermined processing is exhausted from the process chamber **29** by a gas exhaust structure (which is a gas exhauster or a gas exhaust system) (not shown). In the present disclosure, the term “process gas” refers to a gas (or gases) supplied into the process chamber **29**. The same also applies to the following description.

[0065] After a predetermined process time has elapsed, the inert gas is supplied through the gas supply structure, an inner atmosphere of the process chamber **29** is replaced with the inert gas, and the inner pressure of the process chamber **29** is returned to the normal pressure. In the present specification, the term “process time” refers to a time duration of continuously performing a process related thereto. The same also applies to the following description.

<Substrate Unloading Step S4>

[0066] The boat **26** is lowered by the boat elevator **32** through the seal cap **34**. The wafers **18** (which are processed) are unloaded out of the boat **26** in an order reverse to that of loading the wafers **18** into the boat **26** as described above, and then accommodated in the pod **9** of the loading port structure **21**.

<Undocking step S5>

[0067] The opening **44** of the pod **9** accommodating the wafers **18** is closed (or sealed) with the lid **52** by the pod opener **14**, and the pod **9** is undocked by releasing a contact between the pod **9** and the surface **40***a* of the edge of the opening **40***c* by the locking structure **41** and the latching structure **43**.

[0068] Then, the pod **9** accommodating the wafers **18** (which are processed) is unloaded out of the substrate processing apparatus **1** in an order reverse to that of loading the pod **9** as described above.

(4) OTHER EMBODIMENTS (MODIFIED EXAMPLES)

[0069] Subsequently, modified examples of the latching structure **43** in the embodiments mentioned above will be described in detail. In the following modified examples, portions (features) different from those of the embodiments mentioned above will be described in detail.

First Modified Example

[0070] FIG. **7**A is a diagram schematically illustrating a state in which the pod **9** is fixed to the surface **40***a* of the frame **40** by a latching structure **71** according to the present modified example. FIG. **7**B is a diagram schematically illustrating a state in which the pod **9** is released from the surface **40***a* of the frame **40** by the latching structure **71**.

[0071] For example, the latching structure **71** includes: a pair of heads **72** configured to respectively come into contact with at least two recesses **55** formed on each side surface of the flange **54**; a pair of drivers (driving structures) **73** configured to respectively drive the pair of heads **72** to linearly move in the Y-axis direction, that is, in the direction substantially parallel to the surface **40***a*; and a guide structure **74** configured to guide a linear movement of the pair of heads **72**. Hereinafter, each of the pair of heads **72** may also be referred to as a “head **72**”, and each of the pair of drivers **73** may also be referred to as a “driver **73**”. At a tip (front end) of the head **72**, a tapered surface **72***a* is provided. The tapered surface **72***a* is inclined such that a distance from the surface **40***a* increases as it approaches the tip (front end) of head **72**. The driver **73** and the guide structure **74** are fixed to the support **60**. As the driver **73**, for example, an air cylinder capable of moving the head **72** in the Y-axis direction due to a change in an air pressure may be used. The pair of heads **72** is configured to press the flange **54** from both sides by the pair of drivers **73**, respectively, in a synchronized manner with an accuracy enough to prevent the positional deviation in the Y-axis direction of the pod **9**.

[0072] That is, the latching structure **71** is configured to be movable in a direction substantially parallel to the surface **40***a*, and is further configured such that the pair of heads **72** respectively comes into contact with the recesses **55** of the flange **54** to press the pod **9** from both sides without generating the force of pressing the flange **54** against the surface **40***a*.

[0073] According to the present modified example, it is possible to obtain substantially the same effects as those of the embodiments mentioned above. In addition, it is possible to easily set the pressing force such that the pod **9** is contacted while suppressing a deformation of the pod **9**. It is also possible to reduce a vibration occurring when the pod **9** is docked.

Second Modified Example

[0074] FIG. **8**A is a diagram schematically illustrating a state in which the pod **9** is fixed to the surface **40***a* of the frame **40** by a latching structure **81** according to the present modified example.

FIG. **8**B is a diagram schematically illustrating a state in which the pod **9** is released from the surface **40***a* of the frame **40** by the latching structure **81**. FIG. **9** is a diagram schematically illustrating a vertical cross-section of a diaphragm among a pair of diaphragms **83** constituting the latching structure **81**.

[0075] For example, the latching structure **81** includes: a pair of heads **82** configured to respectively come into contact with each side surface of the flange **54**; the pair of diaphragms **83** configured to respectively drive the pair of heads **82** in the Y-axis direction by a working fluid; a pair of housings **84** configured to respectively hold (or support) the pair of diaphragms **83** and configured such that the working fluid can be supplied into each of the pair of housings **84**; and a pair of regulators **85** configured to communicate with an inside (inner portion) of each of the pair of the housings **84** via a piping and configured to adjust a pressure of the working fluid. Hereinafter, each of the pair of heads **82** may also be referred to as a "head **82**", each of the pair of diaphragms **83** may also be referred to as a "diaphragm **83**", each of the pair of housings **84** may also be referred to as a "housing **84**", and each of the pair of regulators **85** may also be referred to as a "regulator **85**". Configurations of the pair of heads **82** are substantially the same with each other, configurations of the pair of diaphragms **83** are substantially the same, with each other and configurations of the pair of housings **84** are substantially the same with each other. In addition, the pair of heads **82** (the pair of diaphragms **83** and the pair of housings **84**) is arranged symmetrically with respect to a line at positions facing each other with the opening **40***c* of the surface **40***a* interposed therebetween. Therefore, in the following description, the present modified example will be described using the head **82**, the diaphragm **83** and the housing **84** which come into contact with the flange **54** on the side surface of the pod **9**.

[0076] As shown in FIG. **9**, a working fluid supply passage **84***a* is provided in the housing **84**. The working fluid supplied through the regulator **85** is filled in the diaphragm **83**. For example, the diaphragm **83** is made of an elastic material such as polyurethane. In addition, a space S is provided inside the diaphragm **83**. The working fluid is filled in the space S. On the other hand, for example, the head **82** is made of a resin (such as polyamide) which is abrasion-resistant and self-lubricating.

[0077] The regulator **85** is configured to set a force with which the head **82** presses the flange **54** to a predetermined value. The regulator **85** may include a three-way valve configured to switch between supplying and discharging the working fluid whose pressure is regulated. By changing the pressure of the working fluid in the head **82** via the diaphragm **83** by using the regulator **85**, it is possible to move the head **82** in the Y-axis direction. For example, the diaphragm **83** and the housing **84** are commercially available as a clamp module, and may also be referred to as a "diaphragm type actuator".

[0078] That is, by increasing the pressure of the working fluid in the space S by using the pair of regulators **85**, the pair of heads **82** is moved in a direction to come into contact with the flange **54**, and the pair of heads **82** presses the flange **54** from both sides. In addition, by decreasing the pressure of the working fluid in the space S by using the pair of regulators **85**, the pair of heads **82** is moved in a direction away from the flange **54**. The latching structure **81** is configured such that, even when the pod **9** without the recess **55** is used, the pair of heads **82** comes into contact with the flange **54** to press the pod **9** from both sides due to a flexibility of the diaphragm **83**.

[0079] According to the present modified example, it is possible to obtain substantially the same effects as those of the embodiments mentioned above. In addition, according to the present modified example, by using the diaphragm type actuator mentioned above, the head **82** can come into contact with the flange **54** in accordance with a shape of the flange **54**. In other words, the head **82** can be used with respect to a variety of types of the pods **9** to be contacted. In addition, by using the diaphragm type actuator, it is possible to prevent an air leakage and a dust generation.

Third Modified Example

[0080] FIG. **10** is a diagram schematically illustrating a latching structure **91** according to the present modified example. In FIG. **10**, in order to explain a configuration of the latching structure

91, the handle **56** provided on the upper surface of the pod **9** is not illustrated. For example, the latching structure **91** is constituted by: a head **82** configured to come into contact with an upper surface of the flange **54**; a driver (driving structure) **93** configured to drive the head **82** in the up-down direction by a working fluid; and a regulator **85** configured to set the force with which the head **82** presses the flange **54** to a predetermined value by adjusting a pressure of the working fluid of the driver **93**. By changing the pressure of the working fluid in the head **82** via the driver **93** by using the regulator **85**, it is possible to move the head **82** in the Z-axis direction and in a Z-axis direction opposite to the Z-axis direction.

[0081] The latching structure **91** is provided at a support **92** provided on the surface **40***a* above the opening **40***c* of the frame **40**. In other words, when the pod **9** is placed on the loading port structure **21**, the latching structure **91** is disposed above the flange **54** of the pod **9**.

[0082] The latching structure **91** is configured to increase the pressure of the working fluid of the driver **93** by using the regulator **85**. Thereby, it is possible to move the head **82** in the −Z-axis direction (which is a direction toward the flange **54**) such that the head **82** comes into contact with the flange **54**. In addition, the latching structure **91** is further configured to decrease the pressure of the working fluid of the driver **93** by using the regulator **85**. Thereby, it is possible to move the head **82** in the Z-axis direction (which is a direction away from the flange **54**) such that a contact between the head **82** and the flange **54** is released. In other words, the latching structure **91** is configured to be capable of linearly moving the head **82** in the Z-axis direction (also referred to as the “up-down direction”).

[0083] According to the present modified example, it is possible to obtain substantially the same effects as those of the embodiments mentioned above. In addition, according to the present modified example, by using the diaphragm type actuator mentioned above, the head **82** can come into contact with the flange **54** in accordance with the shape of the flange **54**. In addition, by using the diaphragm type actuator, it is possible to prevent the air leakage and the dust generation. Further, since the vibration occurring when the pod **9** is docked or undocked can be reduced, it is possible to smoothly perform the operations.

Fourth Modified Example

[0084] FIG. **11** is a diagram schematically illustrating a latching structure **101** according to the present modified example. In FIG. **11**, in order to explain a configuration of the latching structure **101**, the handle **56** provided on the upper surface of the pod **9** is not illustrated. FIG. **12**A is a diagram schematically illustrating a cross-section of the latching structure **101**, taken along a line A-A shown in FIG. **11**, when the latching structure **101** is in contact with the flange **54**, and FIG. **12**B is a diagram schematically illustrating the cross-section of the latching structure **101**, taken along the line A-A shown in FIG. **11**, when the latching structure **101** is released from contact with the flange **54**.

[0085] Similar to the third modified example mentioned above, the latching structure **101** is provided at the support **92** provided on the surface **40***a* above the opening **40***c* of the frame **40**. In other words, when the pod **9** is placed on the loading port structure **21**, the latching structure **101** is disposed above the flange **54** of the pod **9**.

[0086] For example, the latching structure **101** is constituted by: a head **102** configured to come into contact with the upper surface of the flange **54**; and a linear mover (which is a linear moving structure) **103** configured to support the head **102** such that the head **102** can make a linear movement in the Z-axis direction. For example, the linear mover **103** includes: a housing **106**; a bias structure **105** configured to bias the head **102** toward the flange **54** inside the housing **106**; and a support **104** whose one end (first end) is disposed outside the housing **106** and whose the other end (second end) is connected to the head **102**. The bias structure **105** is provided around the support **104** inside the housing **106**, and the bias structure **105** and the support **104** are provided in a coaxial manner. For example, the bias structure **105** is configured as a compression coil spring, and the latching structure **101** is configured to come into contact with and to press the head **102** against the flange **54** by using a restoring force of the bias structure **105** alone. A tapered surface **102***a* is provided at the head **102** to push the head **102** upward while being rubbed against an edge of the flange **54** when the pod **9** is moved from the undocking position to the docking position. A length of the head **102** in the Y-axis direction in the present modified example is set to be sufficiently greater than a distance between ribs of the flange **54** in the Y-axis direction. Thereby, the head **102** can slide smoothly along ridges of the ribs without getting caught in recesses between the ribs.

[0087] According to the present modified example, it is possible to obtain substantially the same effects as those of the embodiments mentioned above. In addition, according to the present modified example, by further using the bias structure **105** with different biasing forces, it is possible to easily set the pressing force such that the pod **9** is contacted while suppressing the deformation of the pod **9**.

[0088] The configurations of the locking structure **41**, the latching structures and the pod **9** described in the embodiments or the modified examples mentioned above are merely examples, and may be modified in accordance with circumstances without departing from the scope of the technique of the present disclosure.

[0089] Further, the process flow described in the embodiments mentioned above is merely an example, and a redundant step may be deleted, a new step may be added or process procedures may be changed, without departing from the scope of the technique of the present disclosure.

[0090] Further, in addition to or instead of a substrate processing apparatus capable of manufacturing a semiconductor device, the technique of the present disclosure may also be applied to other substrate processing apparatuses such as an LCD (Liquid Crystal Display) manufacturing apparatus capable of processing a glass substrate. The contents of the substrate processing may include not only a film forming process of forming a film such as a CVD (Chemical Vapor Deposition) film, a PVD (Physical Vapor Deposition) film, an epitaxial growth film, an oxide film, a nitride film and a metal-containing film, but also a process such as an annealing process, an oxidation process, a diffusion process, an etching process, an exposure process, a photolithography process, a coating process, a molding process, a developing process, a dicing process, a wire bonding process and an inspection process.

[0091] For example, the embodiments mentioned above are described by way of an example in which a vertical type (batch type) substrate processing apparatus capable of simultaneously processing a plurality of substrates is used to process the substrate. However, the technique of the present disclosure is not limited thereto. For example, the technique of the present disclosure may be preferably applied when a single wafer type substrate processing apparatus capable of processing one or several substrates at a time is used to process the substrate. For example, the embodiments mentioned above are described by way of an example in which a substrate processing apparatus including a hot wall type process furnace is used to process the substrate. However, the technique of the present disclosure is not limited thereto. For example, the technique of the present disclosure may be preferably applied when a substrate processing apparatus including a cold wall type process furnace is used to process the substrate.

[0092] The process procedures and the process conditions of each process using the substrate processing apparatuses exemplified above may be substantially the same as those of the embodiments or modified examples mentioned above. Even in such a case, it is possible to obtain substantially the same effects as in the embodiments or the modified examples mentioned above.

[0093] For example, the technique of the present disclosure may also be applied when the embodiments and the modified examples mentioned above are appropriately combined. For example, the process procedures and the process conditions of each process of such a combination may be substantially the same as those of the embodiments or the modified examples mentioned above.

[0094] According to some embodiments of the present disclosure, it is possible to suppress the tilt of the substrate container when the lid of the substrate container is opened.

Claims

1. A loading port structure comprising: a frame constituting a part of a wall configured to separate an inside and an outside of a local clean environment, and comprising: a first surface provided outside the local clean environment and configured to be capable of making an airtight contact with an opening of a substrate container; and an entrance through which a substrate accommodated in the substrate container is capable of passing; and a latching structure provided adjacent to the first surface and configured to be capable of being switched between an engaged state in which the latching structure engages with a flange provided around the opening and a disengaged state in which the latching structure does not engage with the flange so that a movement of the substrate container is unrestricted in an up-down direction and in a front-rear direction, wherein the latching structure comprises: a pair of heads configured to respectively engage with at least two recesses formed on each side of the flange; and a pair of drivers configured to respectively move the pair of heads in a direction substantially parallel to the first surface, wherein each of the pair of heads is provided with a contact portion configured to generate a force of pressing the flange against the first surface by transitioning the latching structure from the disengaged state to the engaged state, and wherein the contact portion comprises a tapered surface.
2. A loading port structure comprising: a frame constituting a part of a wall configured to separate an inside and an outside of a local clean environment, and comprising: a first surface provided outside the local clean environment and configured to be capable of making an airtight contact with an opening of a substrate container; and an entrance through which a substrate accommodated in the substrate container is capable of passing; a locking structure configured to engage with an engagement groove provided in a bottom surface of the substrate container; and a latching structure provided adjacent to the first surface and configured to be movable in a direction substantially parallel to the first surface, wherein the latching structure is further configured to come into contact with a flange provided around the opening and to press the substrate container from both sides thereof or an upper surface thereof without generating a force of pressing the flange against the first surface.
3. The loading port structure of claim 1, wherein the latching structure is not adjacent to other surfaces of the frame than the first surface.
4. The loading port structure of claim 1, wherein the latching structure is further configured to prevent the substrate container from tilting in the front-rear direction even when a pressure inside the local clean environment is 200 Pa or more higher than a pressure outside the local clean environment.
5. The loading port structure of claim 2, wherein the latching structure is further configured to prevent the substrate container from tilting in the front-rear direction even when a pressure inside the local clean environment is 200 Pa or more higher than a pressure outside the local clean environment.
6. The loading port structure of claim 1, further comprising a support table on which the substrate container is placed, and configured to be movable in a direction substantially perpendicular to the first surface.
7. The loading port structure of claim 1, wherein a thickness of the latching structure protruding from the first surface is set to be smaller than a distance between the flange and the first surface at an undocking position where the substrate container is separated from the first surface.
8. The loading port structure of claim 2, wherein a thickness of the latching structure protruding from the first surface is set to be smaller than a distance between the flange and the first surface at an undocking position where the substrate container is separated from the first surface.

9. The loading port structure of claim 1, wherein each of the pair of heads is configured to transition the latching structure between the engaged state and the disengaged state in a synchronized manner with an accuracy enough to prevent a positional deviation in a left-right direction of the substrate container.
10. The loading port structure of claim 1, wherein the pair of drivers are configured to respectively move the pair of heads in a direction without a component in the front-rear direction.
11. The loading port structure of claim 1, wherein each of the pair of heads is supported rotatably and movably in a rotational axis direction, and wherein each of the pair of drivers comprises: a cylinder configured to expand and contract substantially parallel to the flange to be locked with; and a bias structure configured to bias each of the pair of heads in the rotational axis direction.
12. The loading port structure of claim 1, wherein a width of each of the pair of heads is set to 45 mm or less.
13. The loading port structure of claim 2, wherein the latching structure comprises: a head configured to come into contact with the flange; and a bias structure configured to support the head such that the head is capable of making a linear movement, and wherein the latching structure is further configured to press the head against the flange by using a restoring force of the bias structure alone.
14. The loading port structure of claim 12, wherein each of the pair of heads is provided with a tapered surface inclined relative to a direction along which the substrate container moves to a docking position at which the substrate container is pressed against the first surface, and wherein the tapered surface is configured to slide on the flange when the substrate container moves to the docking position such that each head is pushed in a direction opposite to a direction along which the flange is pressed.
15. The loading port structure of claim 2, wherein the latching structure comprises: a head provided above the entrance and configured to come into contact with the flange provided on an upper surface of the substrate container; an actuator configured to drive the head in an up-down direction by a working fluid; and a regulator configured to set the a force with which the head presses the flange to a predetermined value by adjusting a pressure of the working fluid of the actuator.
16. The loading port structure of claim 2, wherein the latching structure comprises: a head configured to come in contact with the flange; and a diaphragm type actuator configured to move the head, wherein the head is attached to the diaphragm type actuator such that the head is movable in a direction opposite to a movement direction thereof.
17. A substrate processing apparatus comprising the loading port structure of claim 1.
18. A substrate processing apparatus comprising the loading port structure of claim 2.
19. A substrate processing method comprising: (a) bringing an opening of a substrate container into an airtight contact with a first surface of a frame, wherein the frame constitutes a part of a wall configured to separate an inside and an outside of a local clean environment and is provided with an entrance through which a substrate accommodated in the substrate container is capable of passing, and wherein the first surface is provided outside the local clean environment; (b) switching a latching structure between an engaged state in which the latching structure engages with a flange provided around the opening and a disengaged state in which the latching structure does not engage with the flange so that a movement of the substrate container is unrestricted in an up-down direction and in a front-rear direction, wherein the latching structure is provided adjacent to the first surface and comprises: a pair of heads configured to respectively engage with at least two recesses formed on each side of the flange; and a pair of drivers configured to respectively move the pair of heads in a direction substantially parallel to the first surface; and (c) processing the substrate taken out from the substrate container through the opening, wherein, in (b), a contact portion in each of the pair of heads is configured to generate a force of pressing the flange against the first surface by transitioning the latching structure from the disengaged state to the engaged state, and wherein the contact portion comprises a tapered surface.

20. A method of manufacturing a semiconductor device, comprising the substrate processing method of claim **19**.